SUNFLASER
OPTOELECTRONIC SOLUTIONS

SFL3300AW

Laser Rangefinder Module

Product Manual

Version 202608V1

Disclaimer

This product manual is intended to provide reference specifications and performance parameters of the product. If certain specifications are not included in this document, please contact SunFlaser Technology (Chengdu) Co., Ltd. directly for further confirmation. All contents of this manual and the specifications of the ranging module are subject to change without prior notice.

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At the time of publication, all descriptions, specifications, designs, and related procedures contained herein are considered valid. SunFlaser Technology (Chengdu) Co., Ltd. reserves the right to modify the above information at any time without prior notice. If you have any questions regarding the use of this manual, please contact us promptly.

Safety Instructions

For the safe use of the ranging module, please carefully follow the safety instructions provided in this manual. Do not attempt to disassemble, modify, or repair any part of the ranging module. Do not tamper with or attempt to alter the performance of the ranging module.

Intended Use

- Take precautions against electrostatic discharge (ESD). Do not touch any electronic components or parts of the ranging module without proper protective measures.
- Keep this manual readily accessible to all personnel operating the ranging module.
- Operate the ranging module only within the specified voltage and power range.
- Operate the ranging module in a clean, dry, and ESD-protected environment.
- Do not touch the optical glass lens with your fingers.
- Avoid exposing the ranging module to environments with drastic temperature fluctuations.

Improper Use

- Using the ranging module with the host system before verifying that all functions are operating properly;
- Operating the ranging module without first reading and understanding this manual and all safety instructions;
- Disassembling, modifying, or altering the ranging module without authorization from SunFlaser Technology (Chengdu) Co., Ltd.;
- Sending operating commands other than those specified in this manual;
- Using or testing the ranging module in non-standard, abnormal, or unsuitable environments;

- Operating the ranging module in a manner inconsistent with the instructions provided in this manual;
- Continuing to operate the ranging module after improper handling or misuse;
- Using a ranging module with obvious faults, defects, or damage;
- Performing test operations within the blind zone of the ranging module.

Improper use may result in the following risks:

- Eye injury
- Incorrect results
- Measurement errors
- Property damage
- Equipment malfunction

Operator Responsibilities

- Be familiar with the basic knowledge required for the correct operation of the ranging module;
- Understand and comply with the operation and maintenance requirements of the ranging module;
- Prevent damage to the ranging module during storage and transportation;
- Keep the ranging module clean and well maintained;
- Ensure that the ranging module is stored and operated in a safe environment;
- Do not expose the ranging module to strong mechanical shock or impact;
- Do not expose the ranging module to environments with drastic temperature fluctuations.

Avoid Measurement Errors

- Pay attention to all factors that may affect the performance of the ranging module;

- After the Laser Rangefinder Module has been subjected to improper handling (such as vibration, impact, or dropping), or before performing critical measurement tasks, it is recommended to verify the performance specifications of the ranging module.

Table of Contents

01 Product Overview

The SFL3300AW Laser Rangefinder Module is independently developed by our company and incorporates a 1535 nm erbium-glass laser. It utilizes a single-pulse time-of-flight (TOF) ranging method.

Developed as a high-precision ranging solution specifically for UAV applications, this series of laser rangefinder modules sets new industry standards through advanced technology. It provides accurate spatial perception capabilities for applications including gimbals, electro-optical platforms, vehicle-mounted systems, and airborne platforms, supporting the intelligent upgrading of integrated equipment. By integrating this module, the target acquisition efficiency for small UAV-type targets can be effectively improved.

02 Key Features

- Single ranging and continuous ranging
- Front and Rear Target Indication
- Range gating

- Command self-test function

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Key Specifications

No.	Parameter	Specification
1	Eye Safety	Class 1
2	Wavelength	1535 ± 5 nm
3	Receiving Aperture	Φ 65 mm
4	Beam Divergence	1mrad±0.2mrad
5	Maximum Range	≥2.1km against a DJI Mavic 3 Pro UAV with dimensions of 0.1 m × 0.2 m, or an equivalent target
		≥3.3 km against a DJI Phantom 4 UAV with dimensions of 0.2 m × 0.3 m, or an equivalent target
6	Minimum Range	≤ 90 m
7	Ranging Accuracy	≤ ±2 m
8	Ranging Frequency	1–10 Hz
9	Range Resolution	≤30m
10	Measurement Success Rate	≥ 98%
11	False Alarm Probability	≤ 1%
12	Communication Interface	RS422
13	Supply Voltage	DC 5–28 V
14	Average Power Consumption	≤1.5 W @ 5 V (at 1 Hz); ≤3 W @ 5 V (at 5 Hz); ≤4 W @ 5 V (at 10 Hz)
15	Peak Power Consumption	≤6 W
16	Standby Power Consumption	≤200 mW

17	Dimensions	≤134 mm×114 mm×80 mm
18	Weight	≤378 g
19	Operating Temperature	−40 °C to +70 °C
20	Storage Temperature	−55 °C to +75 °C
21	Shock Resistance	Tested in accordance with GJB150.18A-2009, Environmental Test Methods for Military Equipment — Shock Test.
22	Vibration Resistance	Tested in accordance with GJB150.16A-2009, Environmental Test Methods for Military Equipment — Vibration Test.

Note:
The specified performance is based on line-of-sight conditions with visibility not less than 20 km, relative humidity ≤ 80%, and target reflectivity ≥ 0.3.

04 Interface Description

Communication Interface: RS-422, baud rate: 115200 bps (default).
Electrical Interface: Connector receptacle model: J30J-9ZKP; mating plug model: J30J-9T. The interface pin assignments are shown in the table below:

Pin No.	Definition	Remarks
1	RS-422 GND	Communication Ground
2	RS-422 RXD+	Laser Serial Receive Positive (Laser-Side Definition)
3	RS-422 RXD-	Laser Serial Receive Negative (Laser-Side Definition)
4	RS-422 TXD-	Laser Serial Transmit Negative (Laser-Side Definition)
5	RS-422 TXD+	Laser Serial Transmit Positive (Laser-Side Definition)

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6	VIN+	Power Input Positive
7	VIN+	Power Input Positive
8	GND_P	Power Input Negative
9	GND_P	Power Input Negative

Pin Sequence: The connector terminals are numbered sequentially from 1 to 9, from top to bottom.

05 Communication Protocol

*** See Appendix 1.**

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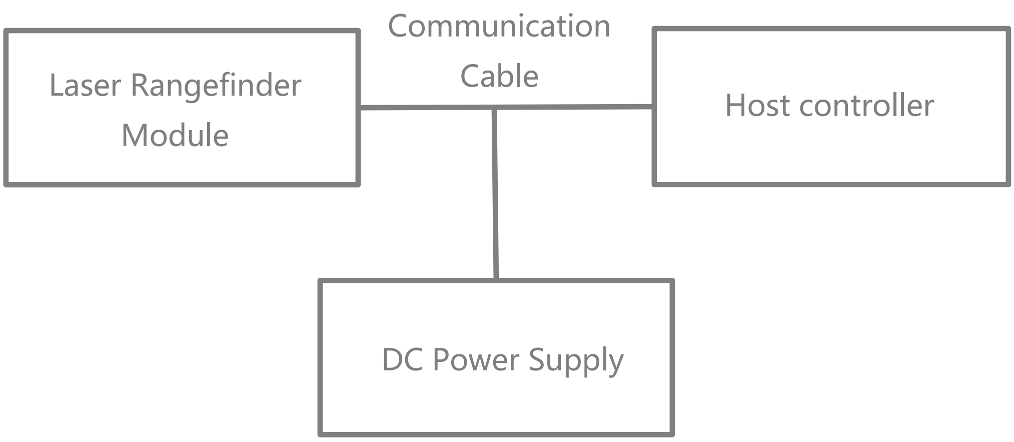
Operating Instructions

Power-On Procedure

- Before powering on, connect the Laser Rangefinder Module, communication cable, DC power supply, and Host Controller according to the electrical interface requirements and as shown in the figure below.



- The module powers on automatically after the power supply is connected.

Power-Off Procedure

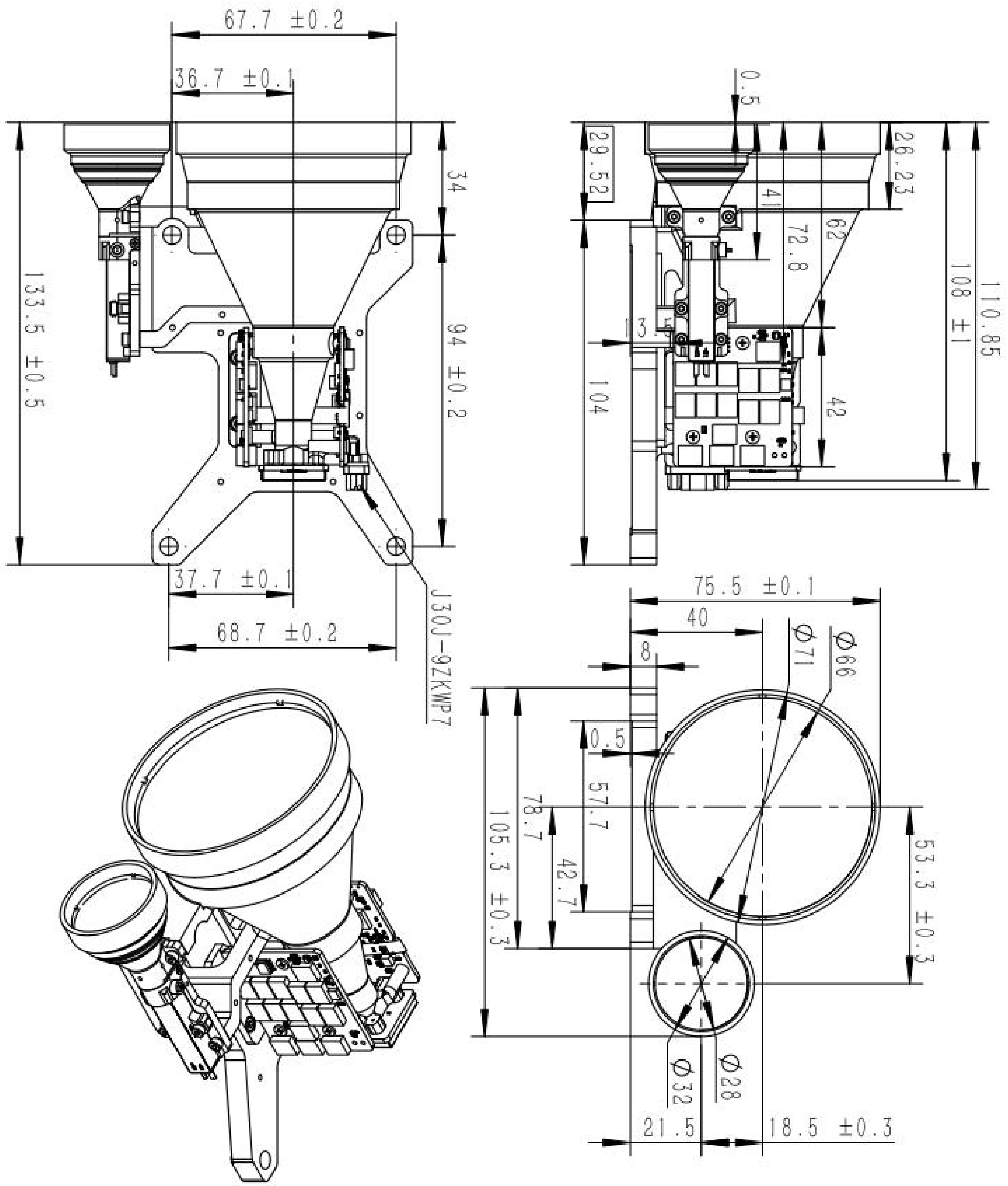
- Before powering off, ensure that all operating processes and tasks have been completed and that the program has been exited properly.
- After confirmation, disconnect the power supply to power off the module normally.

Ranging Mode Operation

- Send the "Single Ranging" command to the Laser Rangefinder Module. The module performs a single ranging operation and returns the ranging status and distance value.
- Send the "Continuous Ranging" command to the Laser Rangefinder Module. The module performs continuous ranging and returns the ranging status and distance value continuously.
- Send the "Stop Ranging" command to terminate the ranging operation.

07

Mechanical Dimensions and Mounting Hole Layout



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Precautions for Use

- The laser emitted by this ranging module is a 1535 nm eye-safe laser. Although it is classified as eye-safe, direct exposure to the laser beam is not recommended;
- **When adjusting the parallelism of the three optical axes, be sure to block the receiving lens. Otherwise, excessive echo signals may permanently damage the detector;**
- This ranging module is not hermetically sealed. Ensure that the relative humidity of the operating environment remains below 80%, and keep the environment clean to prevent damage to the laser device;
- The ranging capability of the module is affected by atmospheric visibility and target characteristics. Fog, rain, and dusty conditions will reduce the ranging distance. Targets with high reflectivity, such as green foliage, white walls, and exposed limestone, can increase the ranging distance. In addition, increasing the inclination angle between the target and the laser beam will reduce the ranging distance;
- **Never direct the laser toward highly reflective targets such as glass or white walls within 30 meters, as excessively strong echo signals may damage the APD detector;**
- Do not connect or disconnect cables while the power is on;
- Ensure correct power supply polarity at all times. Incorrect polarity may result in permanent damage to the equipment.

09 Optical Window Requirements

- The projected size of the effective optical window aperture shall be at least 1.5 mm larger than the outer diameter of the transmitting and receiving lenses of the ranging module;
- The effective aperture of the optical window shall be at least 2 mm larger than the outer diameter of the optical window;
- Since the optical window absorbs a certain amount of laser energy, it is recommended that the thickness of the optical window be controlled within 2–4 mm according to the mechanical dimensions;
- The air gap d between the optical window and the ranging module should be kept as small as possible. A spacing of ≤ 4 mm is recommended;
- For the 1535 nm Laser Rangefinder Module, it is recommended that the optical window be coated with an anti-reflection coating optimized for 1525–1545 nm, with a transmittance of ≥ 99%.

10 Safety Precautions

- To ensure the safety of both operators and test subjects, the following safety measures have been incorporated into the design of the SFL3300AW Laser Rangefinder Module;
- Safety design and analysis are conducted in accordance with GJB 900A-2012, General Requirements for Equipment Safety;

- Non-flammable materials are used, and all mechanical and electrical interfaces are securely connected;
- Appropriate design measures are implemented to prevent moisture accumulation that could result in short circuits;
- The module operates below the safety voltage limit for the human body.

11 Maintenance Instructions

Packing List

The standard items supplied with the product are listed in the table below. Optional accessories may not be included in this list and shall be subject to the actual product delivered.

No.	Item	Quantity	Remarks
1	Laser Rangefinder Module	1 unit	-
2	Certificate of Conformity	1 copy	-
3	Packaging Box	1 pc	-
4	Connector Plug	1 pc	-

Module Cleaning

Cleaning of Optical Components

- Dust particles should be removed using an air blower;
- Fingerprints should first be cleaned with absorbent cotton lightly moistened with a small amount of alcohol-ether mixture, and then wiped with a clean lens cloth.

Cleaning of Mechanical and Electronic Components

- With the power disconnected, gently wipe mechanical and electronic components with alcohol and allow them to dry naturally before use;
- Keep the ranging module, connectors, and cables away from moisture and contaminants whenever possible;
- Ensure that the equipment is completely dry before packaging.

Inspection and Maintenance

Routine Maintenance

When the product is used for the first time, or after replacement of related modules or components, both visual inspection and power-on inspection shall be performed. For products under normal operation, only a power-on inspection is required before use.

- Visual Inspection Procedure
- Check whether the product appearance is normal;
- Check whether the cable connections are correct and securely connected.
- Power-On Inspection Procedure
- Perform the power-on procedure and verify that the standby current is correct;
- Start the self-test function of the product;
- After the inspection is completed, perform the power-off procedure.

Periodic Maintenance

Under normal operating conditions, the Laser Rangefinder Module does not require routine maintenance. If the module has been stored in a dust-free environment for more than one year, maintenance should be performed, including general inspection and power-on inspection.

- General Inspection (Performed with Power Off)
- All markings and serial numbers on the product and test cable connectors shall be correct and clearly legible;
- All screws on the panel shall be securely fastened;

- The optical glass of the product shall be free from visible defects that may affect normal operation, such as spots, pitting, water stains, fungus, fingerprints, dust particles, or cracks.

- Comprehensive Power-On Inspection and Maintenance

- Connect the product power supply in sequence;
- Perform the power-on procedure and verify that the standby current is correct;
- Start the product self-test and complete the self-test procedure;
- Perform the power-off procedure after completion of the inspection.

Troubleshooting

The following list only includes possible fault phenomena not caused by the product itself. If any other technical problems occur during operation, please contact our company for technical support.

Problem / Symptom	Possible Cause	Troubleshooting Method	Diagnostic Criteria	Solution
No communication or abnormal communication with the ranging module	Power supply issue	Check whether the power supply has hardware or software wake-up enabled, verify voltage settings, and ensure the current protection setting is correct. If the voltage is correct, try replacing the power supply.	If the module operates normally after changing the voltage or power supply, the fault is determined to be caused by the power supply.	Set the correct voltage/current or replace the power supply.
	Cable connection issue	Check whether the wiring is secure or replace the communication cable (TTL communication cables should not be excessively long).	If communication is restored after reconnecting or replacing the cable, the fault is determined to be caused by the cable connection.	Reconnect or replace the communication cable.
		Check whether the wiring matches the interface definition.	If the module operates normally after correcting the wiring, the fault is determined to be caused by incorrect wiring.	Rewire according to the correct interface definition.

	Incorrect baud rate setting	1. Check whether the module status and baud rate settings match the communication protocol. 2. If baud rate configuration is supported, set the required baud rate and test communication again.	If communication becomes normal after correcting the baud rate, the fault is determined to be caused by an incorrect baud rate setting.	Set the Host Controller baud rate according to the communication protocol.
	Incorrect communication command	Check whether the sent commands match the communication protocol and confirm that commands are transmitted in hexadecimal format.	If communication becomes normal after sending correct commands, the fault is determined to be caused by incorrect communication commands.	Send the correct commands to the ranging module.
	RS422 driver not installed on Host Controller	Install the RS422 driver on the Host Controller.	If communication becomes normal after driver installation, the fault is determined to be caused by the missing driver.	Install the required driver.
	Main control program issue	If communication fails during integration with the main controller, verify whether standalone communication is normal.	If standalone communication is normal, the fault may be caused by the main control program.	Check the system software, including checksum settings and Host Controller data parsing/display functions.
	Other causes (circuit fault, etc.)	Return to the manufacturer for inspection.	To be determined by factory inspection.	Return for repair.
Invalid ranging data returned	Operational issue	Check whether the transmitting or receiving lens is blocked, and ensure that there are no obstructions in the ranging environment (such as window glass or wall corners).	If ranging returns to normal after removing the obstruction, the fault is determined to be caused by blockage.	Remove the obstruction.
		Check whether the ranging axis and sighting axis are aligned.	If ranging returns to normal after realignment, the fault is determined to be caused by axis misalignment.	Realign the optical axes.

<table>
<tr><td rowspan="2"></td><td>Laser failure</td><td>Check whether the operating current is normal, observe whether the current fluctuates during ranging, send a self-test command, or use an infrared test card/CCD camera to observe whether a laser spot is present.</td><td>If the self-test indicates abnormal laser output and no spot is observed, the fault is preliminarily determined to be caused by laser failure.</td><td>Return for repair.</td></tr>
<tr><td>Other causes (circuit fault, detector issue, etc.)</td><td>Return to the manufacturer for inspection.</td><td>To be determined by factory inspection.</td><td>Return for repair.</td></tr>
<tr><td rowspan="3">False alarm during ranging</td><td>Electrical Noise Interference</td><td colspan="2" rowspan="3">Block the receiving lens of the laser rangefinder module or point it toward the sky (with the module facing away from the sun), and check whether false ranging events still occur. If no false ranging events occur, the issue is determined to be caused by optical noise interference. Further determine whether the interference is caused by external ambient light or by the optical window of the integrated system, such as a mismatch between the optical window transmission band and the operating wavelength of the laser rangefinder module, an excessive optical window tilt angle, an insufficient optical window aperture causing beam obstruction, or excessive contamination of the optical window.

Power and communicate with the laser rangefinder module independently within the integrated system, and check whether false ranging events still occur. If no false ranging events occur, the issue is determined to be caused by electrical or communication noise interference.

Power the laser rangefinder module independently within the integrated system and check whether false ranging events still occur. If no false ranging events occur, the issue is determined to be caused by power supply noise interference.

Establish an independent communication connection with the laser rangefinder module within the integrated system and check whether false ranging events still occur. If no false ranging events occur, the issue is determined to be caused by communication noise</td><td>It is recommended to mitigate power supply noise by using a larger common-mode filter in series, replacing the grounding capacitor with a 10 nF capacitor, and increasing the threshold setting (which may reduce ranging performance).</td></tr>
<tr><td>Communication Noise Interference</td><td>It is recommended to mitigate communication noise by installing a shielding ferrite core around the communication cable or increasing the threshold setting (which may reduce ranging performance).</td></tr>
<tr><td>Optical Noise Interference</td><td>It is recommended to mitigate optical noise by optimizing the optical window and test angle, or by increasing the threshold setting (which may reduce</td></tr>
</table>

		interference. If false ranging events still occur after completing the above procedures, test the laser rangefinder module as a standalone unit outside the integrated system, using an independent power supply and communication connection. If false ranging events still occur, the issue is determined to be related to the laser rangefinder module or the power supply configuration.		ranging performance).
	Other Causes (e.g., Circuit Failures)	Return to the Manufacturer for Inspection	Factory Return Inspection and Evaluation	Return to the Manufacturer for Repair
Insufficient ranging capability	Weather conditions	Check whether weather conditions such as rain, snow, fog, dust, low visibility, high humidity, or atmospheric turbulence are present.	If ranging capability meets specifications under proper weather conditions, the fault is determined to be caused by environmental conditions.	Test under suitable weather conditions.
	Optical window issue	Check whether the optical window has low transmittance, incorrect wavelength coating, contamination, small aperture, or excessive tilt angle. Remove the optical window and retest.	If ranging capability returns to normal after removing the optical window, the fault is determined to be caused by the optical window.	Replace the optical window according to the recommended specifications and installation requirements.
	Target size too small or reflectivity too low	Replace the target with a larger or higher-reflectivity object, preferably perpendicular to the optical axis.	If ranging capability improves with a more suitable target, the fault is determined to be caused by improper target selection.	Use a more suitable target or select a higher-performance ranging module.
	Operating temperature out of range	Send a self-test command and check the module operating temperature. Retest after returning to normal temperature.	If ranging capability returns to normal after temperature recovery, the fault is determined to be caused by excessive operating temperature.	Operate within the specified temperature range.
	Optical axis deviation	Confirm whether the module has undergone vibration, shock,	To be determined by factory inspection.	Return for repair.

		temperature cycling, or drop events.		
	Other causes (circuit fault, low receiver sensitivity, etc.)	Return to the manufacturer for inspection.	To be determined by factory inspection.	Return for repair.
Accuracy out of tolerance	Large target reflectivity deviation	Confirm whether the test target has excessively high reflectivity and whether the returned data fluctuates significantly.	If accuracy returns to normal after changing the target, the fault is determined to be caused by target reflectivity.	Replace the target. High-reflectivity targets are not recommended.
	Improper test operation	Check whether the target is correctly aligned.	If accuracy returns to normal after proper alignment, the fault is determined to be caused by improper operation.	Realign the target.
	Calibration instrument issue	Verify the accuracy of the calibration instrument.	If accuracy returns to normal after replacing the calibration instrument, the fault is determined to be caused by inaccurate calibration equipment.	Replace the calibration instrument.
	Low optical window transmittance	Check whether the returned ranging value is excessively large. Remove the optical window and retest.	If accuracy returns to normal after removing the optical window, the fault is determined to be caused by low optical window transmittance.	Replace the optical window according to the recommended specifications and installation requirements.

Packaging and Transportation Requirements

Packaging

After unpacking, if the product needs to be placed back into storage, it shall be repackaged using the original packaging whenever possible.

If the product needs to be returned to the manufacturer, the original packaging is recommended. If alternative packaging is used, it shall not cause any degradation in product performance or physical damage.

Transportation

Repackaged products may be transported by automobile, railway, aircraft, or ship. During transportation, the package shall be securely fixed to the transport vehicle to avoid impact, improper handling, and exposure to rain or snow.

12 Technical Support and Service

If a product malfunction occurs and factory return is required for fault analysis, troubleshooting, or repair, SunFlaser Technology (Chengdu) Co., Ltd. provides a one-year warranty and lifetime technical support from the date of product delivery.

During the warranty period, product failures caused by manufacturing or quality issues will be repaired or replaced free of charge by SunFlaser. For product damage caused by improper operation or other user-related factors, repair and replacement costs will be charged according to the actual situation.

This product is a precision optical instrument. Please handle and operate it carefully during use. If you have any additional questions regarding operation or maintenance, please contact our after-sales support personnel at any time.

13 Contact Information

Website: www.sunflaser.com

Tel: +86 186 2827 7862 (Sales Manager: Mr. Cao)

Address: No. 58, Zhenxing West 1st Road, Building 1, Room 201, Jinniu District, Chengdu, Sichuan, China

Appendix 1：

Default baud rate: 115200 bps;

Data format: 8 data bits, 1 start bit, 1 stop bit, no parity check. The data packet consists of a header byte, command field, data length field, parameter field, and checksum byte;

Communication mode: The Host Controller and the ranging module use a master-slave communication mode. The Host Controller sends control commands to the ranging module, and the ranging module receives and executes the commands. During ranging operation, the ranging module periodically returns ranging data and status information to the Host Controller according to the ranging cycle. The communication format and command definitions are shown in the following tables.

a) Commands Sent by the Host Controller

The transmission packet format is as follows:

STX0	CMD	LEN	DATA1H	DATA1L	CHK

Table 2. Transmit Packet Format Description

No.	Field Name	Description	Code	Remarks
1	STX0	Start-of-frame identifier	55(H)	—
2	CMD	Command code	See Table 3	—
3	LEN	Data length	Length of the parameter field	—
4	DATAH	Parameter data	See Table 3	High byte
5	DATAL	Parameter data		Low byte
6	CHK	XOR checksum	XOR of all bytes except the checksum byte	—

Command descriptions are provided below:

Table 3. Host Controller Command and Data Definitions for the Ranging Module

No.	CMD	Function	Data Bytes	Remarks	Length	Example Code
1	0x00	Standby (Stop Continuous Ranging)	00 00	Stops ranging operation	6 bytes	55 00 02 00 00 57
2	0x01	Single Ranging	00 00	—	6 bytes	55 01 02 00 00 56
3	0x02	Continuous Ranging	DATAH = XX(H) DATAL = YY(L)	DATA represents the ranging period in ms	6 bytes	55 02 02 03 E8 BE (1 Hz ranging)
4	0x03	Self-Test	00 00	—	6 bytes	55 03 02 00 00 54
5	0x04	Minimum Distance Gate Setting	DATAH = XX(H) DATAL = YY(L)	DATA represents the blind-zone distance in meters	6 bytes	55 04 02 00 64 37 (Minimum distance: 100 m)
6	0x06	Accumulated Laser Emission Count Query	00 00	Query accumulated laser emission count	6 bytes	55 06 02 00 00 51
7	0x22	Target Mode Setting	DATAH = XX(H) DATAL = YY(L)	Single target: 00 00 Triple target: 00 01 First/Last target: 00 10	6 bytes	55 22 02 00 01 74 (Triple target mode)
8	0x26	Baud Rate Setting	DATAH = XX(H) DATAL = YY(L)	The baud rate parameter is equal to the baud rate × 0.01.	6 bytes	55 26 02 04 80 (Baud rate:115200)

b) Host Controller Receive Format

The receive packet format is as follows:

STX0	CMD	LEN	DATAn	DATA0	CHK

Table 4. Receive Packet Format Description

No.	Field Name	Description	Code	Remarks
1	STX0	Start-of-frame identifier	55(H)	—
2	CMD_JG	Data command code	See Table 5	—
3	LEN	Data length	Length of the parameter field	—
4	Dn	Parameter data	See Table 5	High-order data byte(s)
5	D0	Parameter data		Low-order data byte
6	CHK	XOR checksum	XOR of all bytes except the checksum byte	—

Status Definitions for Data Received by the Host Controller:

Table 5. Ranging Module Response Data Definitions

No.	CMD	Function	Data Bytes	Remarks	Total Length
1	0x00	Standby (Stop Continuous Ranging)	00 00	Same as request frame	6 bytes
2	0x01	Single Ranging (In single-target mode, the second and third target values are zero; in first/last target mode, the third target value is zero)	D9 D8 D7 D6 D5 D4 D3 D2 D1 D0	D8–D6: First target distance (unit: 0.1 m) D5–D3: Second target distance (unit: 0.1 m) D2–D0: Third target distance (unit: 0.1 m)Targets are ordered from nearest to farthest. D9 (bit7–bit0) status byte: bit7: Main echo flag; 1 = valid main echo, 0 = no main echo. bit6: Echo flag; 1 = echo detected, 0 = no echo. bit5: Laser status; 1 = normal, 0 = laser fault. bit4: Timeout flag; 1 = normal, 0 = timeout. bit3: Reserved (set to 1). bit2: APD status; 1 = normal, 0 = error. bit1: Front target flag; 1 = front target detected, 0 = no front target detected (targets before the main target are considered front targets). (Modulation bit, for reference only.) bit0: Rear target flag; 1 = rear target detected, 0 = no rear target detected (targets after the main target are considered rear targets). (Modulation bit, for reference only.)	14 bytes

3	0x02	Continuous Ranging (In single-target mode, the second and third target values are zero; in first/last target mode, the third target value is zero)	D9 D8 D7 D6 D5 D4 D3 D2 D1 D0	D8–D6: First target distance (unit: 0.1 m) D5–D3: Second target distance (unit: 0.1 m) D2–D0: Third target distance (unit: 0.1 m) Targets are ordered from nearest to farthest. D9 (bit7–bit0) status byte: bit7: Main echo flag; 1 = valid main echo, 0 = no main echo. bit6: Echo flag; 1 = echo detected, 0 = no echo. bit5: Laser status; 1 = normal, 0 = laser fault. bit4: Timeout flag; 1 = normal, 0 = timeout. bit3: Reserved (set to 1). bit2: APD status; 1 = normal, 0 = error. bit1: Front target flag; 1 = front target detected, 0 = no front target detected (targets before the main target are considered front targets). (Modulation bit, for reference only.) bit0: Rear target flag; 1 = rear target detected, 0 = no rear target detected (targets after the main target are considered rear targets). (Modulation bit, for reference only.)	14 bytes
4	0x03	Self-Test	D7 ~ D0	D7–D6: −5 V supply voltage, unit: 0.01 V. D5–D4: Blind-zone value, unit: 1 m. D3: APD high voltage value, unit: V. D2: char type, representing the APD temperature in °C. D1–D0: +5 V supply voltage, unit: 0.01 V.	12 bytes
5	0x04	Minimum Distance Gate Setting	D1 D0	DATA represents the minimum distance value in meters; high byte first, same as request frame.	6 bytes(Saved after power-off)
6	0x06	Accumulated Laser Emission Count Query	D3 ~ D0	DATA represents the accumulated laser emission count; 4-byte value, high byte first.	8 bytes
7	0x22	Target Mode Setting	D1 D0	Same as transmit frame.	6 bytes
8	0x26	Baud Rate Setting	D1 D0	D1 D0: Factory baud rate parameter, high byte first.	6 bytes

Note:
① Undefined data bytes/bits default to 0.